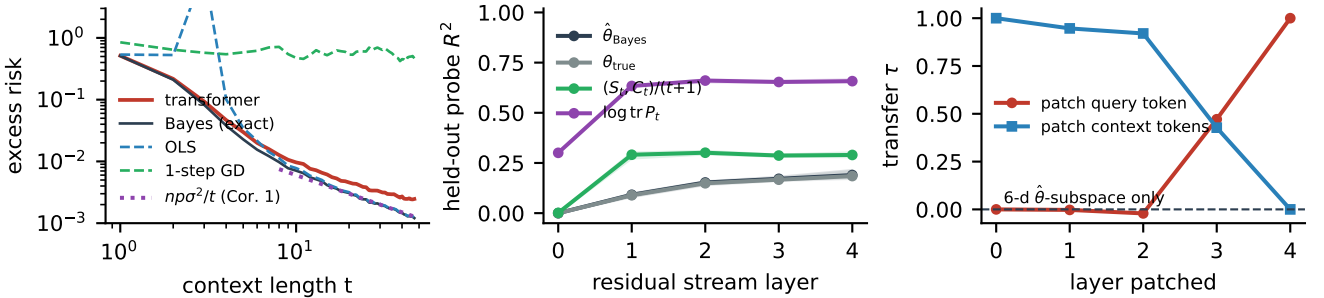
(a) in-context prediction
excess risk
context length t
transformer
Bayes (exact)
OLS
1-step GD
$npσ^2/t$ (Cor. 1)
(b) linear probes
held-out probe R^2
residual stream layer
$\hat{\theta}_{\text{Bayes}}$
θ_{true}
$(S_t, C_t)/(t+1)$
$\log \operatorname{tr} P_t$
(c) activation patching
transfer τ
layer patched
patch query token
patch context tokens
6-d $\hat{\theta}$-subspace only